



# Structure of BMP file

Bitmap file in RAM or ROM contains a header, which should be omitted (we do not check it, assuming that the bitmap already has the required format). After the header there is data section, containing information on pixels' colours. Single byte (8 bits) contains colour information on two pixels, 4 MSB concerns pixel on the left side, 4 LSB - pixel on the right side. Colour is encoded as a 4-bit address in colour table (which can be found in BMP header). The colour table used in the file from this exercise has the following structure (parameter **Bits per pixel=4, NumColors=16**):

| Pixel<br>(hex) | Driving outputs of Spartan-3 Starter Kit board |       |       | Displayed colour |
|----------------|------------------------------------------------|-------|-------|------------------|
|                | red_o                                          | grn_o | blu_o |                  |
| 0              | 0                                              | 0     | 0     | <b>black</b>     |
| 1              | X                                              | X     | X     | <i>unused</i>    |
| 2              | X                                              | X     | X     | <i>unused</i>    |
| 3              | X                                              | X     | X     | <i>unused</i>    |
| 4              | X                                              | X     | X     | <i>unused</i>    |
| 5              | X                                              | X     | X     | <i>unused</i>    |
| 6              | X                                              | X     | X     | <i>unused</i>    |
| 7              | X                                              | X     | X     | <i>unused</i>    |
| 8              | X                                              | X     | X     | <i>unused</i>    |
| 9              | 1                                              | 0     | 0     | <b>red</b>       |
| A              | 0                                              | 1     | 0     | <b>green</b>     |
| B              | 1                                              | 1     | 0     | <b>yellow</b>    |
| C              | 0                                              | 0     | 1     | <b>blue</b>      |
| D              | 1                                              | 0     | 1     | <b>magenta</b>   |
| E              | 0                                              | 1     | 1     | <b>cyan</b>      |
| F              | 1                                              | 1     | 1     | <b>white</b>     |

Fig. 1 Colour table

The order of the pixels in BMP file is as follows: from left to right, from bottom to top (first pixel is from lower left corner of the picture). In the first approach, the picture can be displayed upside down, just to test the reading data from memory.

Each line is filled with zeros at the end, so each line has a length of multiple of 32 bits. In this example filling is not used, since each line has 256 pixels, i.e. exactly 32 groups of 32 bits.

| Basic BMP File Format |                          |                      |                                                                                                                                                                                                                    |                 |
|-----------------------|--------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Name                  |                          | Size                 | Description                                                                                                                                                                                                        |                 |
| Header                |                          | 14 bytes             | Windows Structure: BITMAPFILEHEADER                                                                                                                                                                                |                 |
|                       | Signature                | 2 bytes              | 'BM'                                                                                                                                                                                                               |                 |
|                       | FileSize                 | 4 bytes              | File size in bytes                                                                                                                                                                                                 |                 |
|                       | reserved                 | 4 bytes              | unused (=0)                                                                                                                                                                                                        |                 |
|                       | DataOffset               | 4 bytes              | File offset to Raster Data                                                                                                                                                                                         |                 |
| InfoHeader            |                          | 40 bytes             | Windows Structure: BITMAPINFOHEADER                                                                                                                                                                                |                 |
|                       | Size                     | 4 bytes              | Size of InfoHeader = 40                                                                                                                                                                                            |                 |
|                       | Width                    | 4 bytes              | Bitmap Width                                                                                                                                                                                                       |                 |
|                       | Height                   | 4 bytes              | Bitmap Height                                                                                                                                                                                                      |                 |
|                       | Planes                   | 2 bytes              | Number of Planes (=1)                                                                                                                                                                                              |                 |
|                       | BitCount                 | 2 bytes              | Bits per Pixel<br>1 = monochrome palette. NumColors = 1<br>4 = 4bit palletized. NumColors = 16<br>8 = 8bit palletized. NumColors = 256<br>16 = 16bit RGB. NumColors = 65536 (?)<br>24 = 24bit RGB. NumColors = 16M |                 |
|                       | Compression              | 4 bytes              | Type of Compression<br>0 = BI_RGB no compression<br>1 = BI_RLE8 8bit RLE encoding<br>2 = BI_RLE4 4bit RLE encoding                                                                                                 |                 |
|                       | ImageSize                | 4 bytes              | (compressed) Size of Image<br>It is valid to set this =0 if Compression = 0                                                                                                                                        |                 |
|                       | XpixelsPerM              | 4 bytes              | horizontal resolution: Pixels/meter                                                                                                                                                                                |                 |
|                       | YpixelsPerM              | 4 bytes              | vertical resolution: Pixels/meter                                                                                                                                                                                  |                 |
|                       | ColorsUsed               | 4 bytes              | Number of actually used colors                                                                                                                                                                                     |                 |
|                       | ColorsImportant          | 4 bytes              | Number of important colors<br>0 = all                                                                                                                                                                              |                 |
| ColorTable            |                          | 4 * NumColors bytes  | present only if Info.BitsPerPixel <= 8<br>colors should be ordered by importance                                                                                                                                   |                 |
|                       |                          | Red                  | 1 byte                                                                                                                                                                                                             | Red intensity   |
|                       |                          | Green                | 1 byte                                                                                                                                                                                                             | Green intensity |
|                       |                          | Blue                 | 1 byte                                                                                                                                                                                                             | Blue intensity  |
|                       |                          | reserved             | 1 byte                                                                                                                                                                                                             | unused (=0)     |
|                       | repeated NumColors times |                      |                                                                                                                                                                                                                    |                 |
| Raster Data           |                          | Info.ImageSize bytes | The pixel data                                                                                                                                                                                                     |                 |

Fig. 2 Structure of BMP file